Esports Tournament Database Management System

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DBMS Project

Introduction

Project Goal: To provide a reliable, structured system that simplifies tournament operations while maintaining high data accuracy

- Growth of E-Sports: Competitive gaming is expanding rapidly, making manual tournament oversight increasingly inefficient.
- Operational Challenges: Traditional management methods often lead to data duplication and critical errors in scheduling.
- The Solution: There is a clear need for a centralized, database-driven application to automate participant organization and result tracking.

Abstract

System Overview

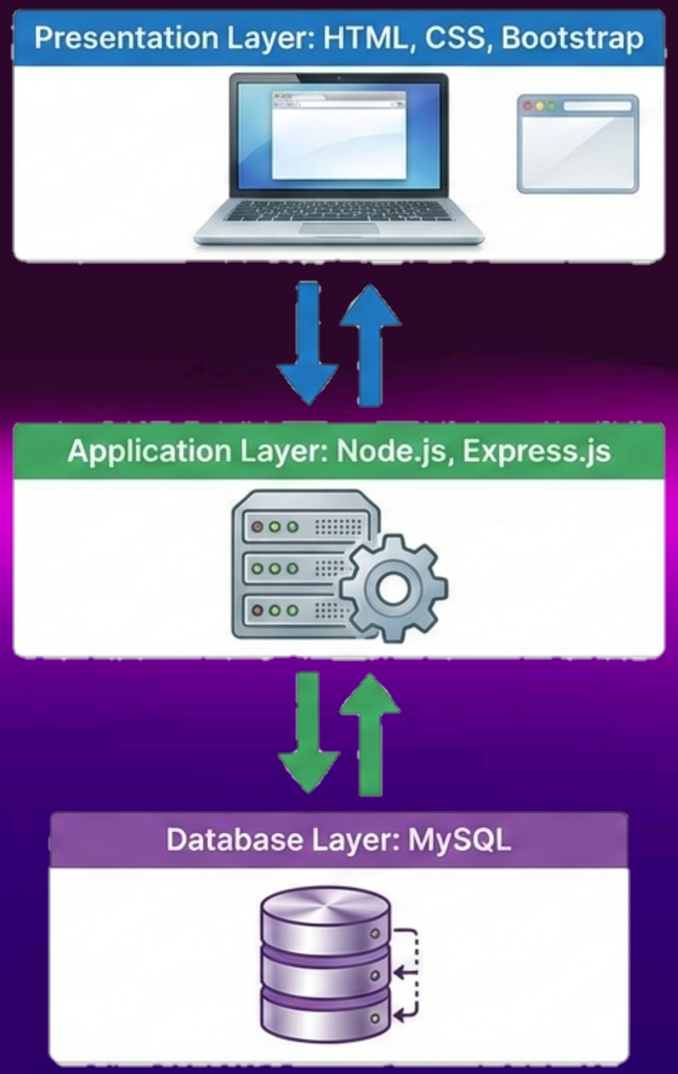
- A full-stack web application built to manage the end-to-end lifecycle of e-sports tournaments.
- Automation: The system automates player registration, team creation, tournament scheduling, and result monitoring.
- User Roles: Includes dedicated interfaces for Administrators (to manage matches) and Players/Teams (to track performance).
- Data Reliability: Uses a structured relational database to ensure that all tournament data is consistent and easily retrievable.

Tech Stack & Rationale

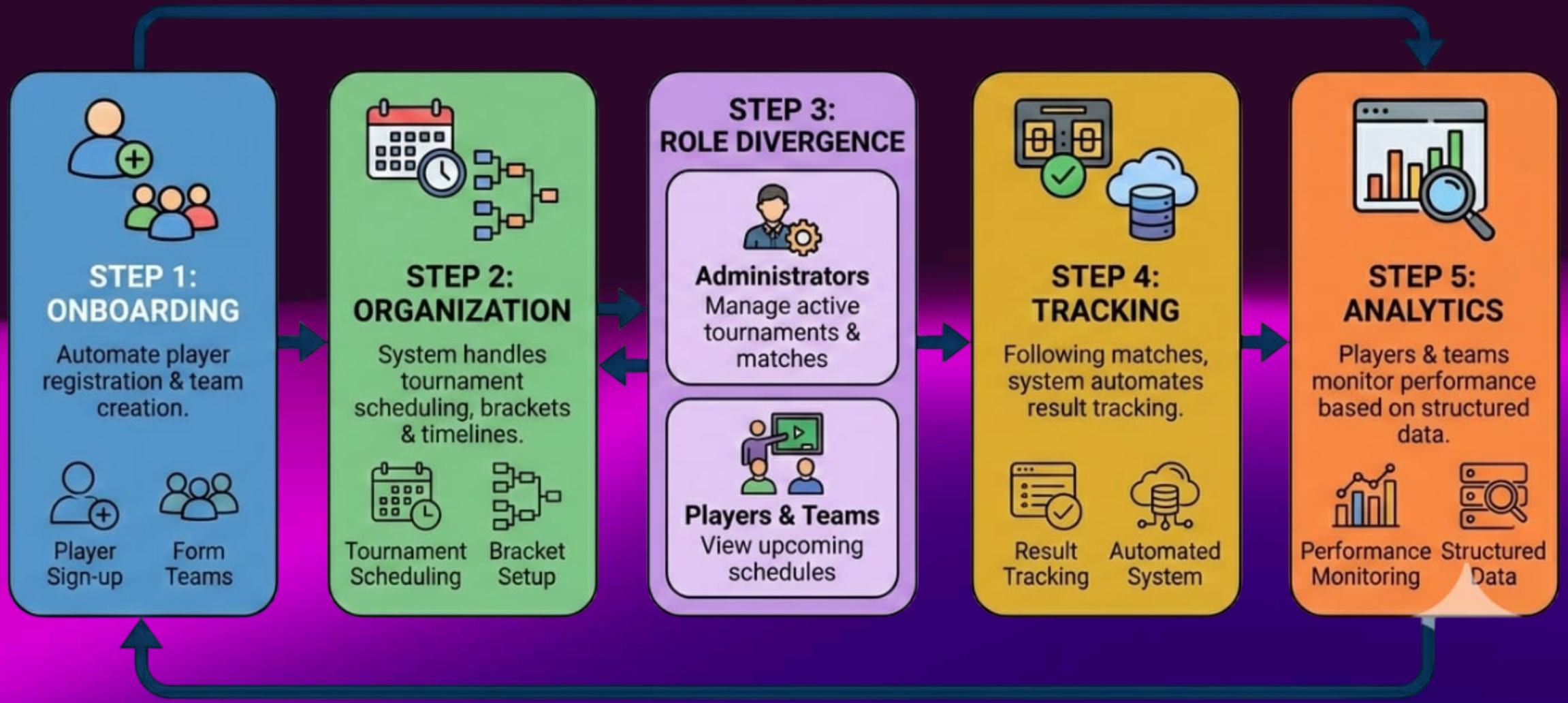
- Frontend: HTML, CSS, & Bootstrap
  - Why: Creates a responsive Presentation Layer. Bootstrap ensures the UI is professional and accessible across various devices used by gamers.
- Backend: Node.js & Express.js
  - Why: Acts as the Application Layer. These technologies are ideal for handling the asynchronous logic required for real-time tournament updates and scheduling automation.
- Database: MySQL
  - Why: Serves as the Database Layer. MySQL is chosen for its ability to handle complex relationships between users, teams, and matches using primary and foreign keys to maintain strict data integrity

Tech Stack Flow Chart

- User Interface: Users interact with the Bootstrap-powered frontend to send requests.
- Server Logic: The Node.js/Express.js backend processes these requests and executes business logic (e.g., generating a match schedule).
- Data Persistence: MySQL stores and retrieves the required data, ensuring that team rosters and match results are linked correctly.



Flow Chart of Description of Project



Thank You